

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 only forms an OSPF adjacency with R1 because its other interfaces are passive. Therefore, R3 learns routes to the 192.168.2.0 network through R1 instead of directly from R2.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router) # no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

____65 (One T1 serial link (at a cost of 64) and the R2 LAN link (at a cost of 1)).

Is R2 listed as an OSPF neighbor to R3? __Yes__

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

____The default reference bandwidth (100 Mbps) makes high-speed links such as Gigabit or 10-Gigabit Ethernet cost the same. Increasing the reference bandwidth allows OSPF to calculate more accurate costs and properly prefer faster links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__OSPF calculates route cost using the sum of the interface costs along the path.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

__1,562, as each serial link costs 781, and the route to the 192.168.23.0/24 network goes through two serial links.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___As OSPF will choose the route with the least cost.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

___As the router ID uniquely identifies each router in the OSPF domain. Controlling it ensures stability in the network, prevents unexpected changes after reloads, and helps maintain consistent OSPF neighbor relationships.

2. Why is the DR/BDR election process not a concern in this lab?

___As most connections between routers are point-to-point serial links, which do not perform DR/BDR elections. DR/BDR elections only occur on broadcast multi-access networks such as Ethernet segments.

3. Why would you want to set an OSPF interface to passive?

___As a passive interface prevents OSPF hello packets and routing updates from being sent on that interface while still advertising the connected network. This reduces unnecessary routing traffic and improves security.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

___As OSPF uses a multi-area hierarchical design with a backbone area.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

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_____ network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

_____ The EIGRP route will be used because it has the lowest administrative distance.
